MATH-18A *Exam 3 Review*

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3.1 Intro to Functions



- **function** $f(x)$ — takes an input, does something to it, and produces *exactly one* output (per input)

Techniques/Skills

- **solve f(#)**: # is *input*, so plug in x = #; simplify
- **solve $f(x)$ = #**: # is *output*, so replace $f(x)$ with the definition of the function; solve for x

3.2 Domain

- **domain** — collection of all function *inputs*
- **range** — collection of all function *outputs*

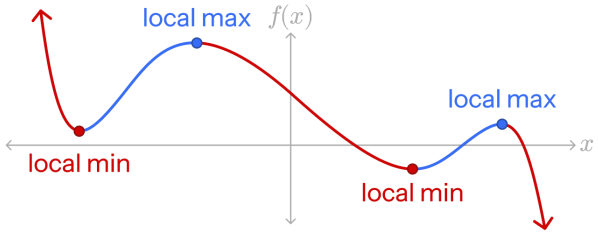
Things to Know

- DANGER ZONES: when denominator is 0; when inside of an even root is negative

Techniques/Skills

- **finding holes/gaps**: identify danger zone; solve for $x \neq$ or $x <$; write answer in interval notation
- **finding domain**: everything EXCEPT holes/gaps

3.3 Graphing Functions



- **increasing** — as inputs increase, outputs ↗
- **decreasing** — as inputs increase, outputs ↘
- **constant** — as inputs increase, outputs stay same
- **relative/local maximum** — point where the function goes from increasing to decreasing (↗↘)
- **relative/local minimum** — point where the function goes from decreasing to increasing (↘↗)

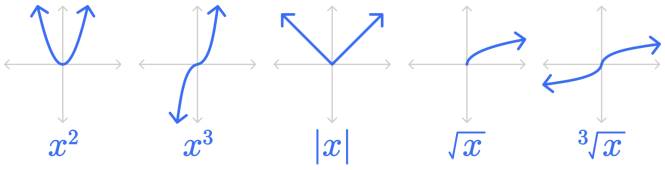
Things to Know

- VERTICAL LINE TEST: checks if function or not

Techniques/Skills

- **get domain & range from graph**: domain is the part of the *x-axis* that the curve covers; range is the part of the *y-axis* that the curve covers

3.4 Transforming Functions



- **transformation** — scaling/shifting a function
- **parent function** — un-transformed function

Things to Know

- TRANSFORMATION FORMULA: $a \cdot f(b \cdot (x-h)) + k$
  $f(\ )$ = parent function
  a = vertical scaling b = horizontal scaling
  k = vertical shift h = horizontal shift

Techniques/Skills

- **formula → graph**: get a, b, h, k from eq.; describe what they do; draw parent function and then draw transformed graph (make sure to scale first!!)
- **graph → formula**: identify a, b, h, k, and parent function; write transformed parent function
- **graphs with unknown parent functions**: apply transformations to key points (vertices, corners)

3.5 Algebra with Functions

- **function composition** $(f \circ g)(x) = f(g(x))$ — inner function nested inside an outer function

Things to Know

- shorthands: $(f+g)(x)$, $(f-g)(x)$, $(fg)(x)$, $(f/g)(x)$
- generally, $(f \circ g)(x) \neq (g \circ f)(x)$ ← order matters!!

Techniques/Skills

- solving function composition: start with inner function; then use inside as INPUT for outside

3.6 Inverse Functions

- **inverse function** $f^{-1}(x)$ — undoes a function $f(x)$
- **one-to-one** — every input has a unique output

Things to Know

- f and f^{-1} SWAP INPUT/OUTPUT: $f(a)=b$ & $f^{-1}(b)=a$
- graph of inverse flips across the diagonal
- HORIZONTAL LINE TEST: checks one-to-one

Techniques/Skills

- **finding inverse**: rename $f(x)$ as y; isolate x; swap all x and y; rename new y as $f^{-1}(x)$
- **checking inverse**: $(f^{-1} \circ f)(x) = x$ *AND* $(f \circ f^{-1})(x) = x$